

Retail Sales Analysis Report

Data Analytics Internship Project
Internship Studio

Field	Details
Project Title	Retail Sales Data Analysis and Reporting
Tools Used	Python, SQL Server, Excel, Jupyter Notebook
Dataset Source	Kaggle - regivm/retailtransactiondata
Total Transactions	125,000
Total Customers	6,889
Date Range	May 2011 - March 2015
Total Revenue	\$8,123,989.00
Avg Transaction	\$64.99

Section 1: Project Overview

The aim of this project is to use Python, SQL Server, and Excel to analyze retail sales data and generate meaningful reports for a retail chain. The project follows a structured 4-phase approach: data collection and database setup, data cleaning and preparation, data analysis, and reporting.

1.1 Dataset Description

The dataset was sourced from Kaggle (regivm/retailtransactiondata) and consists of two CSV files:

- Retail_Data_Transactions.csv: 125,000 rows with fields customer_id, trans_date, tran_amount
- Retail_Data_Response.csv: 6,884 rows with fields customer_id and response (0 = no response, 1 = responded)

1.2 Tools and Technologies

Tool	Purpose
Python (pandas, matplotlib, seaborn)	Data loading, cleaning, analysis, and visualization
SQL Server 2025 / SSMS 22	Relational database setup and data storage
SQLAlchemy + pyodbc	Python to SQL Server connectivity
openpyxl	Automated Excel report generation
Jupyter Notebook (Anaconda)	Interactive development environment

Section 2: Phase 1 - Data Collection and Database Setup

2.1 Data Collection

The dataset was downloaded from Kaggle as two CSV files and stored at C:\Users\Asus\Downloads\archive\. The files were loaded into Python using pandas, validated, and uploaded to SQL Server using SQLAlchemy with Windows Authentication.

2.2 Database and Table Creation

A SQL Server database named retail_sales was created in SSMS 22. Two tables were created using SQL DDL commands:

- transactions: stores customer_id, trans_date, tran_amount with computed columns trans_month and trans_year
- customer_response: stores customer_id and promotion response (0 or 1)

2.3 SQL Server Tables - Verified in SSMS

	transaction_id	customer_id	trans_date	tran_amount	trans_month	trans_year
1	1	CS5295	2013-02-11	35.00	2	2013
2	2	CS4768	2015-03-15	39.00	3	2015
3	3	CS2122	2013-02-26	52.00	2	2013
4	4	CS1217	2011-11-16	99.00	11	2011
5	5	CS1850	2013-11-20	78.00	11	2013

Figure 1: transactions table - TOP 5 rows verified in SSMS

	customer_id	response
1	CS1112	0
2	CS1113	0
3	CS1114	1
4	CS1115	1
5	CS1116	1

Figure 2: customer_response table - TOP 5 rows verified in SSMS

2.4 Data Loading Results

Table	Rows Loaded	Status
transactions	125,000	Successfully uploaded
customer_response	6,884	Successfully uploaded

Clearing duplicate data from transactions table...

Table cleared. Reloading correct data...

transactions : 125,000 rows in DB

Done. Correct row count restored.

Figure 3: Data fix confirmation - 125,000 correct rows restored to SQL Server

Section 3: Phase 2 - Data Cleaning and Preparation

3.1 Missing Values Check

All columns in both tables were checked for missing values using `pandas isnull().sum()`. No missing values were found in any column of either table.

```
Loading data from SQL Server...
transactions : 250,000 rows
response     : 6,884 rows

Missing values in transactions:
transaction_id    0
customer_id       0
trans_date        0
tran_amount       0
trans_month       0
trans_year        0
dtype: int64

Missing values in customer_response:
customer_id    0
response       0
dtype: int64
```

Figure 4: Missing values check - all columns show 0 missing values

3.2 Duplicate Rows and Summary

Both tables were checked for duplicate rows. No duplicates were found. The duplicate check results are shown below along with the missing values confirmation:

```
Missing values in transactions:
transaction_id    0
customer_id       0
trans_date        0
tran_amount       0
trans_month       0
trans_year        0
dtype: int64

Missing values in customer_response:
customer_id    0
response       0
dtype: int64

Duplicate rows in transactions      : 0
Duplicate rows in customer_response: 0
```

Figure 5: Duplicate rows check - 0 duplicates in both tables

3.3 Outlier Detection Using IQR Method

The IQR method was applied to the `tran_amount` column. The lower bound was -7.00 and the upper bound was 137.00. All 125,000 values fell within this range (min: \$10, max: \$105), confirming zero outliers

in the dataset.

```
Outlier Analysis - tran_amount:
count      125000.000000
mean        64.991912
std         22.860006
min         10.000000
25%         47.000000
50%         65.000000
75%         83.000000
max         105.000000
Name: tran_amount, dtype: float64
```

```
IQR Lower Bound : -7.00
IQR Upper Bound : 137.00
Total Outliers  : 0 rows
Outlier %       : 0.00%
```

Figure 6: Outlier analysis - IQR method confirms 0 outliers

Check	transactions	customer_response
Missing Values	0	0
Duplicate Rows	0	0
Outliers (IQR)	0 rows	N/A
Amount Range	\$10.00 - \$105.00	N/A
Mean Amount	\$64.99	N/A

3.4 Calculated Fields Created

The following fields were created per customer for analysis:

- total_spend: Sum of all transaction amounts per customer
- avg_transaction: Average transaction value per customer
- transaction_count: Total number of transactions per customer
- lifespan_days: Days between first and last purchase
- trans_month and trans_year: Extracted from trans_date for time-series analysis

Customer Summary Table:

customer_id	total_spend	avg_transaction	transaction_count	first_purchase	last_purchase	lifespan_days
CS11112	1012	67.47	15	2011-06-15	2015-01-14	1309
CS11113	1490	74.50	20	2011-05-27	2015-02-09	1354
CS11114	1432	75.37	19	2011-07-14	2015-02-12	1309
CS11115	1659	75.41	22	2011-08-10	2015-03-05	1303
CS11116	857	65.92	13	2011-06-27	2014-08-25	1155
CS11117	1185	69.71	17	2011-05-20	2014-07-02	1139
CS11118	1011	67.40	15	2011-05-18	2015-03-14	1396
CS11119	1158	77.20	15	2012-02-28	2015-03-05	1101
CS11120	1677	69.88	24	2011-05-26	2015-03-06	1380
CS11121	1524	58.62	26	2011-05-30	2015-02-03	1345

Total unique customers: 6,889

Monthly Sales Summary:

trans_year	trans_month	total_sales	total_orders	avg_order_value
2011	5	98951	1485	66.63
2011	6	174527	2707	64.47
2011	7	178097	2726	65.33
2011	8	188631	2914	64.73
2011	9	169173	2605	64.94
2011	10	182634	2839	64.33
2011	11	166921	2570	64.95
2011	12	181405	2812	64.51
2012	1	177987	2737	65.03
2012	2	170135	2619	64.96

Figure 7: Customer Summary Table - first 10 customers shown

Monthly Sales Summary:

trans_year	trans_month	total_sales	total_orders	avg_order_value
2011	5	98951	1485	66.63
2011	6	174527	2707	64.47
2011	7	178097	2726	65.33
2011	8	188631	2914	64.73
2011	9	169173	2605	64.94
2011	10	182634	2839	64.33
2011	11	166921	2570	64.95
2011	12	181405	2812	64.51
2012	1	177987	2737	65.03
2012	2	170135	2619	64.96
2012	3	180453	2743	65.79
2012	4	168000	2613	64.29

Total months in dataset: 47

Figure 8: Monthly Sales Summary - first 12 months shown

3.5 Summary Tables Saved to SQL Server

Saving customer_summary to SQL Server...

customer_summary saved : 6,889 rows

Saving monthly_summary to SQL Server...

monthly_summary saved : 47 rows

Verification:

customer_summary in DB : 6,889 rows

monthly_summary in DB : 47 rows

Phase 2 complete. Data is cleaned and prepared for analysis.

Figure 9: Phase 2 completion - both tables saved and verified in SQL Server

Table	Rows	Description
customer_summary	6,889	Per-customer aggregated spend and activity metrics
monthly_summary	47	Monthly aggregated sales figures across 47 months

Section 4: Phase 3 - Data Analysis

4.1 Key Metrics Summary

Metric	Value
Total Transactions	125,000
Total Revenue	\$8,123,989.00
Unique Customers	6,889
Average Transaction Value	\$64.99
Average Customer Spend	\$1,179.27
Promotion Response Rate	9.40% (647 out of 6,884 customers)
Average Customer Lifespan	1,240 days (approximately 3.4 years)
Peak Revenue Year	2013 (\$2,137,368)

4.2 Monthly Sales Trend (2011 - 2015)

Revenue remained consistently between \$160,000 and \$190,000 per month from mid-2011 through 2014. Two dips are visible in early 2013 and early 2014. The sharp drop at the end reflects incomplete 2015 data (January to March only).

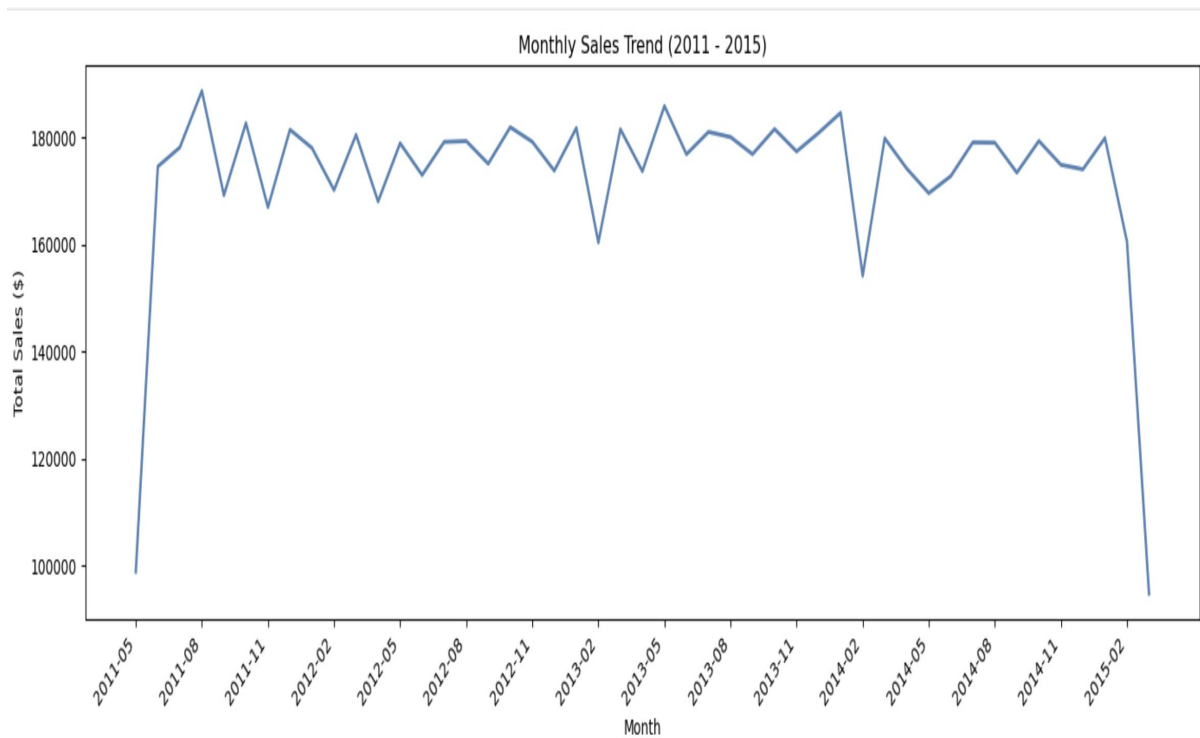


Figure 10: Monthly Sales Trend - May 2011 to March 2015

4.3 Yearly Sales Comparison

Full-year revenue for 2012, 2013, and 2014 was consistently around \$2.1 million, indicating a stable and well-performing retail business.

Year	Total Sales (\$)	Total Orders	Note
2011	1,340,339	20,658	Partial year (May-Dec)
2012	2,116,599	32,541	Full year

2013	2,137,368	32,904	Full year - Peak
2014	2,094,508	32,226	Full year
2015	435,175	6,671	Partial year (Jan-Mar)

Yearly Sales Summary:

trans_year	total_sales	total_orders
2011	1340339.0	20658
2012	2116599.0	32541
2013	2137368.0	32904
2014	2094508.0	32226
2015	435175.0	6671

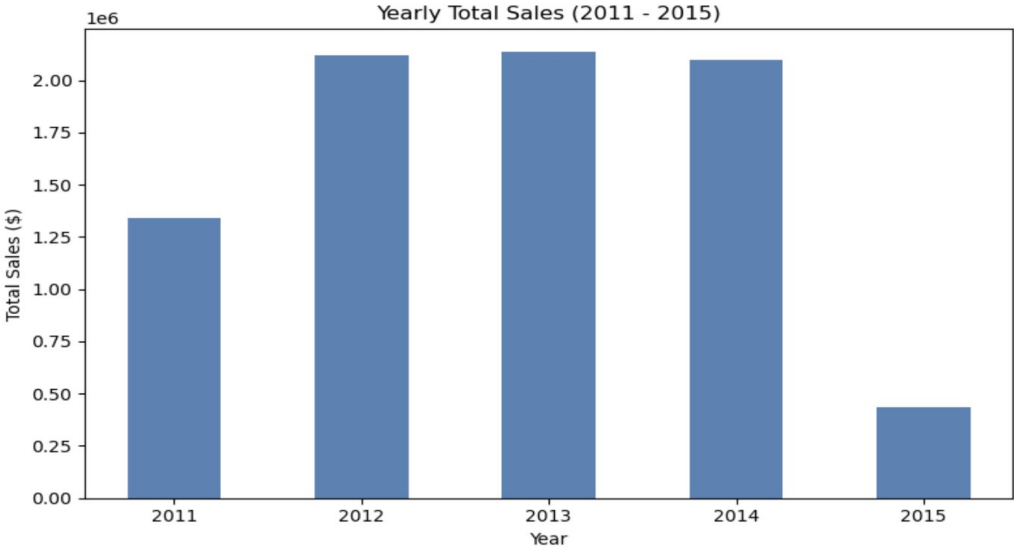


Figure 11: Yearly Total Sales Comparison (2011-2015)

4.4 Top 10 Customers by Total Spend

The top 10 customers all spent between \$2,439 and \$2,933. Customer CS4424 leads with \$2,933 across 39 transactions, reflecting strong customer loyalty and high purchase frequency.

Customer ID	Total Spend (\$)	Transactions	Avg Transaction (\$)
CS4424	2,933	39	75.21
CS4320	2,647	38	69.66
CS5752	2,612	33	79.15
CS4660	2,527	33	76.58
CS3799	2,513	36	69.81
CS5109	2,506	35	71.60
CS4074	2,462	34	72.41
CS3805	2,453	35	70.09
CS4608	2,449	32	76.53
CS5555	2,439	31	78.68

Top 10 Customers by Total Spend:

customer_id	total_spend	transaction_count	avg_transaction
CS4424	2933	39	75.21
CS4320	2647	38	69.66
CS5752	2612	33	79.15
CS4660	2527	33	76.58
CS3799	2513	36	69.81
CS5109	2506	35	71.60
CS4074	2462	34	72.41
CS3805	2453	35	70.09
CS4608	2449	32	76.53
CS5555	2439	31	78.68

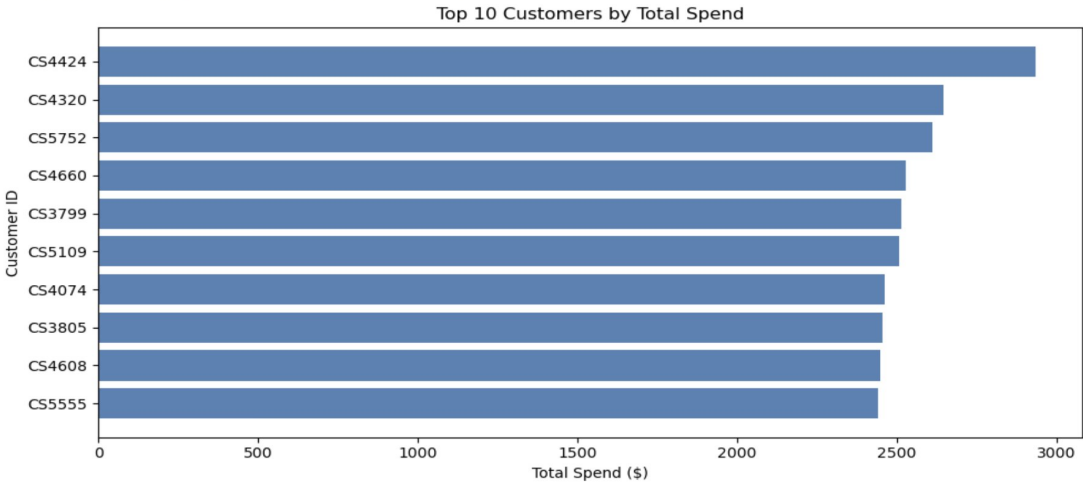


Figure 12: Top 10 Customers by Total Spend

4.5 Customer Promotion Response Analysis

Only 9.4% of customers responded to promotions. However, responding customers spent 28.5% more on average and made 20% more transactions, making them significantly more valuable to the business.

Response Group	Count	Avg Total Spend (\$)	Avg Transaction (\$)	Avg Transactions
Did Not Respond	6,237	1,149.08	62.52	17.82
Responded	647	1,476.89	68.58	21.39

Customer Response Overview:
 Total customers : 6,884
 Responded (1) : 647
 Did not respond (0) : 6,237
 Response rate : 9.40%

Average Spend by Response Group:

response	total_spend	avg_transaction	transaction_count
0	1149.08	62.52	17.82
1	1476.89	68.58	21.39

Customer Promotion Response Rate

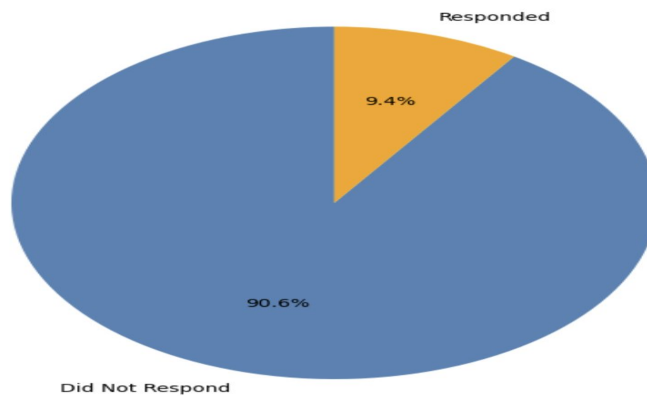


Figure 13: Customer Promotion Response Rate

4.6 Transaction Amount Distribution

Transaction amounts are uniformly distributed between \$10 and \$105 with peaks around \$48 and \$68. The boxplot confirms the median stayed consistently at ~\$65 across all years.

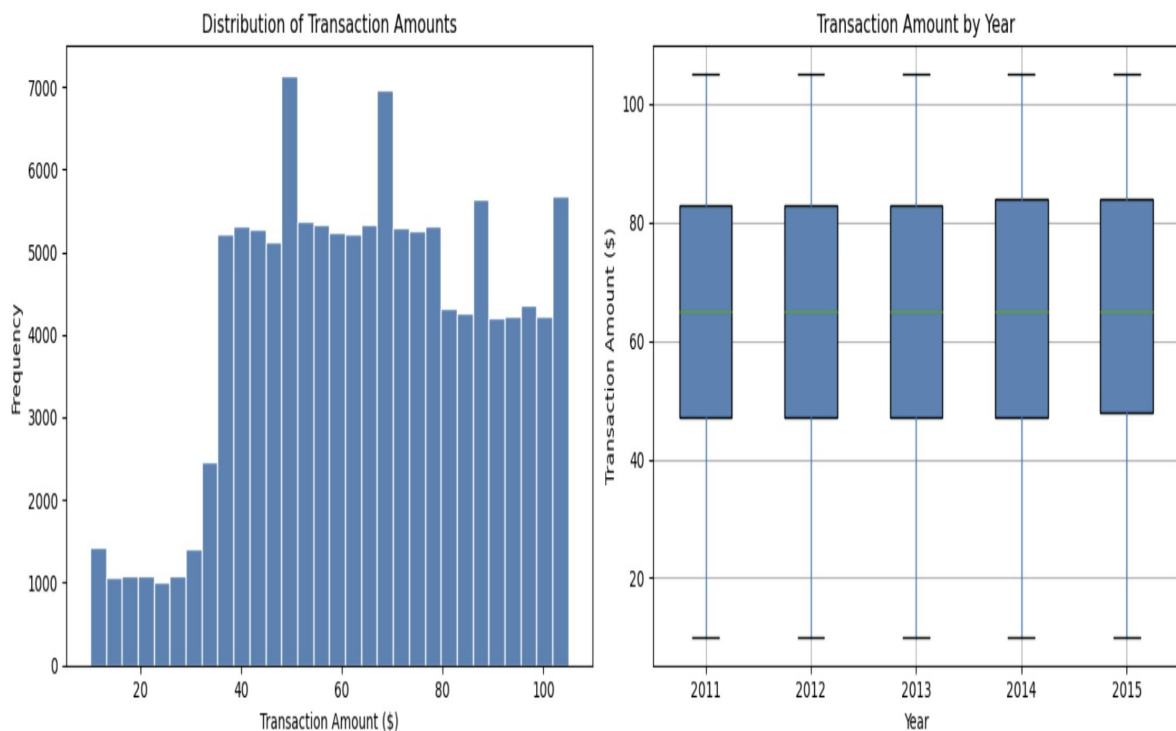


Figure 14: Transaction Distribution Histogram and Yearly Boxplot

4.7 Customer Spend and Transaction Count Distribution

Most customers spent between \$500 and \$1,800 in total (average \$1,179). Most customers made 18-20 transactions, confirming a consistently active and loyal customer base.

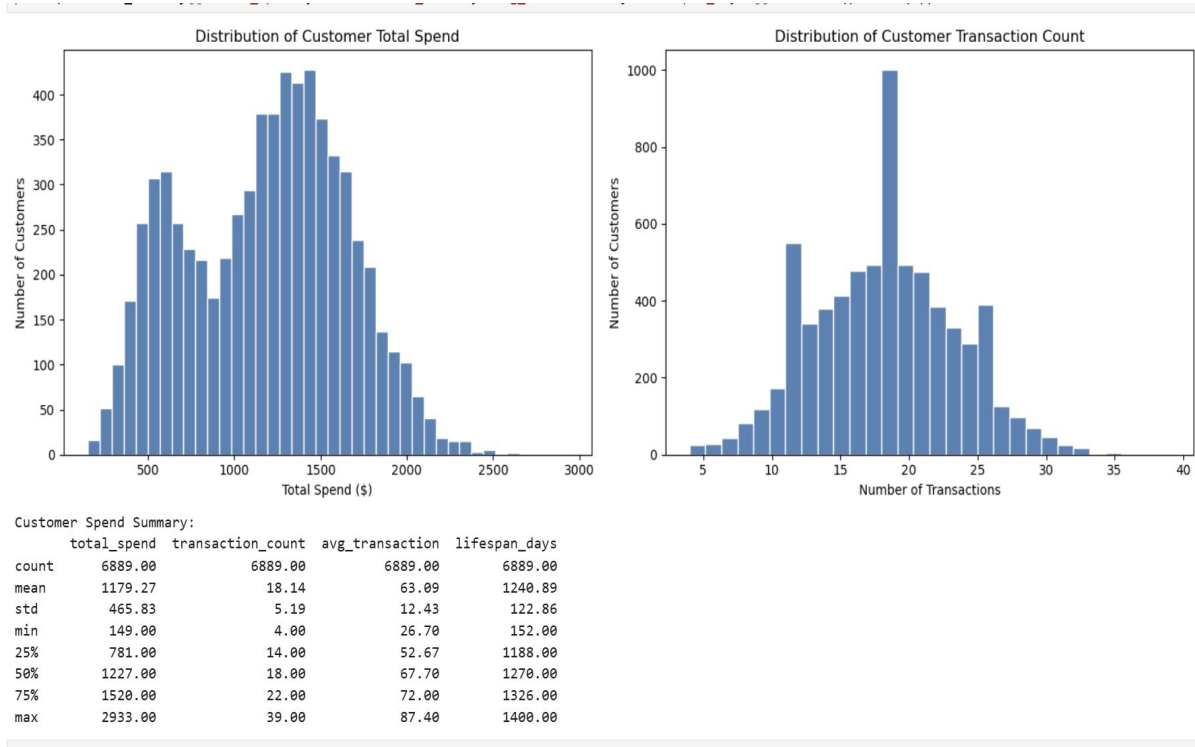


Figure 15: Customer Total Spend and Transaction Count Distribution

Section 5: Phase 4 - Excel Reporting

A professional Excel report was generated using Python's openpyxl library. The report contains 4 sheets providing tabular and visual summaries of all findings.

Sheet	Contents
Executive Summary	8 key business metrics including total revenue, customers, and response rate
Monthly Sales	All 47 months of sales data with total sales, orders, and avg order value
Customer Summary	Top 20 customers by spend plus promotion response group comparison
Charts	All 6 visual analysis charts embedded in the worksheet

5.1 Executive Summary Sheet Data

Metric	Value
Total Transactions	125,000
Total Revenue	\$8,123,989.00
Unique Customers	6,889
Avg Transaction Value	\$64.99
Avg Customer Spend	\$1,179.27
Promotion Response Rate	9.40%
Date Range	May 2011 - March 2015
Total Months	47

5.2 Monthly Sales Data (All 47 Months)

Period	Year	Month	Total Sales (\$)	Orders	Avg Order (\$)
2011-05	2011	5	98,951	1,485	66.63
2011-06	2011	6	174,527	2,707	64.47
2011-07	2011	7	178,097	2,726	65.33
2011-08	2011	8	188,631	2,914	64.73
2011-09	2011	9	169,173	2,605	64.94
2011-10	2011	10	182,634	2,839	64.33
2011-11	2011	11	166,921	2,570	64.95
2011-12	2011	12	181,405	2,812	64.51
2012-01	2012	1	177,987	2,737	65.03
2012-02	2012	2	170,135	2,619	64.96
2012-03	2012	3	180,453	2,743	65.79
2012-04	2012	4	168,000	2,613	64.29
2013-02	2013	2	160,359	2,478	64.71
2014-02	2014	2	154,151	2,375	64.91
2015-03	2015	3	94,829	1,453	65.26

Note: Selected months shown above. Full 47-month data available in the Excel report.

5.3 Top 20 Customers Data

Customer ID	Total Spend (\$)	Avg Transaction (\$)	Transactions	Lifespan (Days)
CS4424	2,933	75.21	39	1,335
CS4320	2,647	69.66	38	1,309
CS5752	2,612	79.15	33	1,317
CS4660	2,527	76.58	33	1,365
CS3799	2,513	69.81	36	1,223
CS5109	2,506	71.60	35	1,328
CS4074	2,462	72.41	34	1,286
CS3805	2,453	70.09	35	1,299
CS4608	2,449	76.53	32	1,348
CS5555	2,439	78.68	31	1,363
CS3163	2,413	77.84	31	1,263
CS2951	2,382	74.44	32	1,199
CS1215	2,362	67.49	35	1,152
CS4540	2,361	73.78	32	1,310
CS2620	2,360	67.43	35	1,256
CS4048	2,358	76.06	31	1,347
CS4998	2,358	69.35	34	1,320
CS1982	2,354	73.56	32	1,296
CS3270	2,349	71.18	33	1,301
CS4746	2,337	75.39	31	1,390

Section 6: Conclusions and Recommendations

6.1 Key Conclusions

- The retail chain generated total revenue of \$8.12 million over 47 months (May 2011 - March 2015).
- Annual revenue for full years 2012-2014 was consistently around \$2.1 million, indicating a stable business.
- The average transaction value of \$64.99 remained consistent across all years, reflecting stable pricing.
- The promotion response rate of 9.4% is low, but responding customers spend 28.5% more and transact 20% more frequently.
- The top 10 customers all spent over \$2,400 in total, highlighting a loyal high-value segment worth nurturing.
- Average customer lifespan of 3.4 years demonstrates strong long-term customer retention.
- Transaction amounts between \$10 and \$105 with zero outliers indicate consistent and fair pricing.

6.2 Recommendations

- Target non-responding customers with personalized promotions since responders show significantly higher lifetime value.
- Investigate the revenue dips in early 2013 and 2014 to identify seasonal patterns and plan inventory accordingly.
- Develop a loyalty rewards program for top customers to maintain and grow their spend over time.
- Expand promotion campaigns given clear evidence that responding customers are more active and valuable.
- Monitor monthly revenue using the automated Excel dashboard for early detection of downturns.
- Consider cohort analysis to understand which customer acquisition periods produce the most loyal customers.

End of Report